Financial Incentives and Facility Births: Evidence from India's Janani Suraksha Yojana

Powell-Jackson, Mazumdar & Mills (2015), *Journal of Health Economics* 43, 154–169

Your Name

Health Economics

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Outline

1. Introduction
2. Background (§2)
3. Data (§3)
4. Empirical strategy (§4)
5. Main results (§5)
6. Heterogeneity & robustness (§6, §5.4)
7. Links to the course
8. Conclusion (§7)

Slido poll • let us start with a poll

In the policies we have studied, does expanding *access* to a health input usually deliver the *health outcome* it was meant to?

A Usually yes — getting people to use a service is the hard part

B Only if the *quality* of the service is also good

C Only if people actually *use* what they are given, not just own it

D Rarely — behaviour and incentives usually undo the gains

Hold that thought. We test it today with one large programme.

Motivation: a recurring theme in this course

A theme runs through the course: a policy can **expand access** to a health input, yet the **health outcome does not follow**.

Our Week 9 paper on **Ujjwala (LPG)**, Gupta et al. (2019), is one case. LPG ownership rose to about three-quarters of rural households, but about three-quarters of them **still burned solid fuel**. Ownership did not become clean-air use.

This paper asks the same kind of question for a **cash incentive** to give birth in a facility.

The pattern to test

subsidy / incentive

↓

more of the input

(facility births, LPG)

↓ ?

better health outcome

(survival, clean air)

The question mark is where this paper does its work.

Slido poll • respond on Slido

An earlier *Lancet* study found large gains from the JSY. What do you think this paper finds for newborn mortality?

A A large fall in deaths, matching the earlier study

B A small but clear fall

C No statistically detectable effect

D Deaths actually rose

The next slide is the answer.

The question

The Janani Suraksha Yojana (JSY) pays women to give birth in a public facility. The expected result is clear. More women use skilled care, so fewer newborns die.

An earlier *Lancet* study supported this. It found facility births up **40+ pp** and neonatal deaths down **2–6 per 1000**.

This paper does not find that result. The design is tighter and the data are larger. The effect on mortality is **not different from zero**. The cash also changes behaviours that the policy did not target.

Three questions

1. Did the cash raise **use** of care?
2. Did use lower **mortality**?
3. What **else** did it change?

Why it is difficult

There is no randomisation. The programme started where need was large. So a simple comparison is confounded.

Context: high mortality, already in decline

- India had approximately **72,000** maternal deaths per year — the largest number of any country
- Maternal mortality fell **47%** (398 to 178 per 100,000) from 1997–98 to 2010–12
- State differences are large: Kerala is approximately 5 times lower than the poorest northern states

Raw means, before and during the JSY (Table 1)

	Before	During
Neonatal mortality /1000	33.0	**26.6**
Facility births (%)	38.7	**43.7**
public	20.0	25.5
private	18.7	18.2
Breastfeeding <1h (%)	31.1	**39.9**

The identification problem

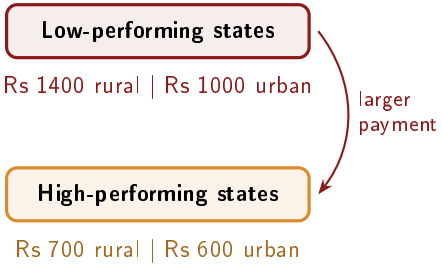
Outcomes improved in all states. The design must separate the JSY effect from this national trend.

The means move in the expected direction. But a correlation is not the causal effect of the JSY.

The programme: a conditional cash transfer (§2.1)

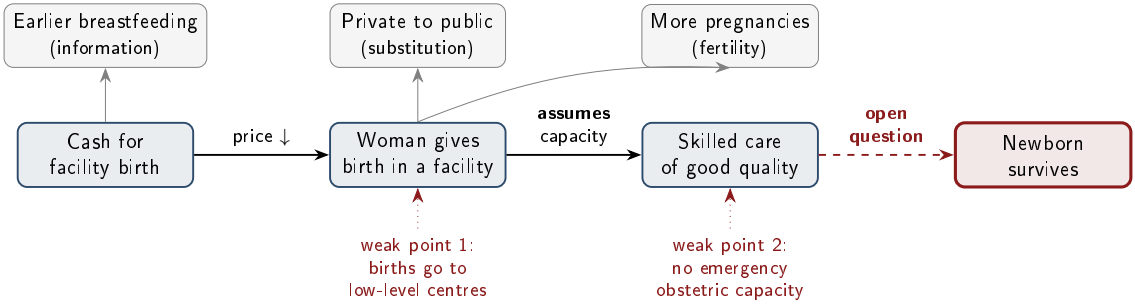
- The programme started in 2005 under the National Rural Health Mission
- It pays cash to a woman who gives birth in a **public** facility (or an accredited private one), within one week
- Community health workers (ASHAs) get **Rs 200–600** for each delivery they bring in — a supply-side payment
- The payment is **larger in low-performing states**

Scale (2007/08): **Rs 8.8 bn** (≈$207m) per year, **7.1m** beneficiaries. A public delivery costs approximately $25.



This asymmetry is important. It returns in the heterogeneity results.

Anticipated effects (§2.2): the causal chain



The cash pays for stage 1. Newborns survive only if stages 2 and 3 also hold.

Anticipated effects (§2.2): what theory predicts

The authors set out the predictions before they look at the data. Standard theory gives one main effect and three side effects.

Main effect: more use

A short-term price cut raises demand for facility care. But the effect on health is uncertain. It depends on the quality of care in the facility, which may be low.

Fertility

Payment per birth is an implicit reward for a birth. It can raise fertility — a shorter interval between births, or more children over a lifetime.

Substitution

The cash is for public facilities only. So some women move from private to public care. This is a transfer between providers, not a net rise in facility use.

Information

More contact with staff can change behaviour at birth, for example earlier breastfeeding or antiseptic cord care.

The study tests all four. The side effects are the paper's contribution: earlier work looked at use and mortality only.

Evidence on the JSY (§2.3): the result this paper revisits

Lim et al. (2010), *Lancet* is the main prior study. It used three identification strategies:

1. **Individual-level matching**
2. **Before-and-after** comparison
3. **District-level difference-in-differences**

The headline results came from strategies 1 and 2. The difference-in-differences result (strategy 3) was weaker and got less emphasis.

Why this paper doubts strategy 1 and 2

- Matching has low power and **reverse selection**: women who give birth in a facility are not a random group
- Before-and-after relies on a strong assumption of **no other change** over the period

This paper uses the stronger design (strategy 3) and finds the mortality effect does not hold.

Data: the District Level Health Survey (§3.1)

Two cross-sections

- DLHS-2 (2002–04): 507,622 women, 593 districts
- DLHS-3 (2007–08): 643,944 women, 611 districts
- **587** districts are in both rounds and form the sample
- Births 2001–2008 (financial years)

Measures

- **Outcomes** from the birth histories: facility use, neonatal and one-day mortality
- **JSY coverage**: the share of public-facility mothers who got the cash, by district and year
- **Unintended outcomes**: pregnancy and breastfeeding (DLHS-3)

Each observation is a woman's most recent delivery (use equations) or a live birth (mortality equations). For pregnancy, the unit is a woman-year, from DLHS-3 only.

The coverage measure, and its weakness (§3.2)

Definition. JSY_{dt} is the number of women who got the cash, as a share of women who gave birth in a public facility in district d, year t.

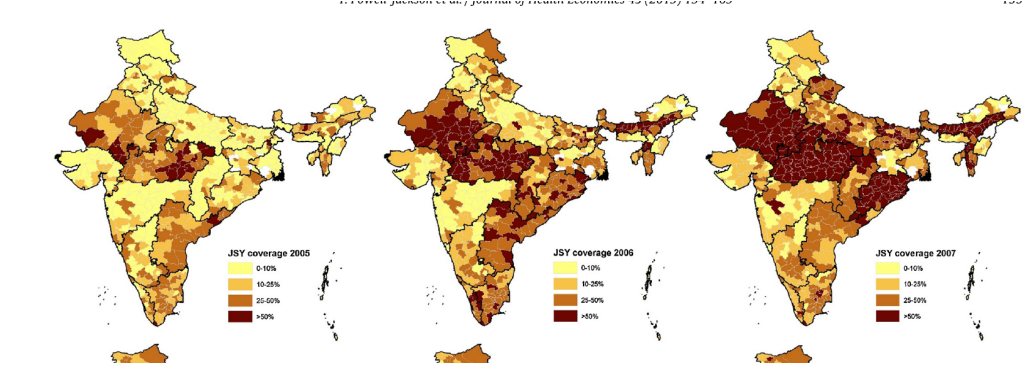
Why a share, not a count. It measures how far the programme reached, by district and year. The authors argue this is a **supply-side** measure, set by administration, not by demand.

The built-in problem

The cash goes to women who give birth in a facility. So coverage is **mechanically linked** to the facility-birth rate: the outcome is in the denominator of the treatment.

This is why the paper leans on district fixed effects and cross-district variation, not on the level of coverage itself.

The variation: staggered district roll-out



From 2005 to 2007 the map gets darker as coverage rises. This timing identifies the effect, after district and year fixed effects.

The threat

The dark districts are in the poor northern belt. Adoption is **not random**. It follows need.

Figure 1 (top row). Powell-Jackson et al. (2015). Darker = higher coverage.

The strategy: difference-in-differences

The authors compare the change in outcomes across districts with different growth in JSY coverage. The method has three steps.

1. A district fixed effect removes everything about a district that does not change over time (its wealth, its geography, its base level of care)
2. A year fixed effect removes any shock common to all districts in a year (the national decline in mortality)
3. What remains is the link between the *rise* in a district's coverage and the *change* in its outcomes

The identifying assumption

Without the JSY, high-coverage and low-coverage districts would have followed **parallel trends**. You cannot test this directly. The paper probes it with pre-trends (§5.4).

The specification, term by term (Eq. 1)

$$y_{idt} = \beta_0 + \beta_1 \,\mathrm{JSY}_{dt} + \vartheta_t Z_d \beta_2 + X_{idt}\beta_3 + \omega_d + \tau_t + \varepsilon_{idt}$$

y_{idt}	Outcome for woman i, district d, year t: facility use, or neonatal/one-day mortality
JSY_{dt}	Coverage, as **three dummies**: 10–25%, 25–50%, >50% (reference: <10%). β_1 is the effect at each level
Z_d	Three district traits: below-poverty-line share, tribal share, mean wealth asset score
$\vartheta_t Z_d$	Those traits **interacted with year** — lets poor and tribal districts follow their own trends
X_{idt}	Household controls: mother's and husband's education, maternal age, wealth, recall period, and dummies for residence, religion, ethnicity, parity, multiple birth, survey round
$\omega_d,\ \tau_t$	District and year **fixed effects**
ε_{idt}	Error, **clustered by district**

Slido poll • respond on Slido

The cash reached districts at different intensities. Where do you expect the effect on facility births?

A It rises smoothly with coverage, at every level

B Only in districts with the highest (>50%) coverage

C Strongest at low coverage, then it flattens

D No effect at any level

Use of care (§5.1, Table 3)

Table 3
Association of JSY with use of maternal health care services.

Dependent variable	Health worker in attendance at delivery	Delivery in a health facility	Delivery in public health facility	Delivery by type of public health facility			At least three ANC visits
				Hospital	Community health centre	Primary health centre	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Baseline model							
JSY coverage 10–25%	−0.0053	−0.0061	−0.00072	0.00081	0.0010	0.0023	0.00071
	(0.0055)	(0.0054)	(0.0049)	(0.0044)	(0.0017)	(0.0019)	(0.0053)
JSY coverage 25–50%	0.0017	0.0072	0.019***	0.0033	0.0092***	0.011***	0.0052
	(0.0061)	(0.0061)	(0.0055)	(0.0047)	(0.0023)	(0.0027)	(0.0061)
JSY coverage >50%	0.056***	0.075***	0.11***	0.028***	0.040***	0.051***	0.010
	(0.0090)	(0.0093)	(0.0086)	(0.0061)	(0.0045)	(0.0039)	(0.0075)
Panel B. Baseline model with district and individual controls							
JSY coverage 10–25%	−0.0021	−0.0033	−0.0015	−0.00011	0.0016	0.0018	0.00099
	(0.0048)	(0.0047)	(0.0047)	(0.0040)	(0.0017)	(0.0020)	(0.0048)
JSY coverage 25–50%	0.0028	0.0075	0.013**	0.00017	0.010***	0.0078***	0.0035
	(0.0053)	(0.0053)	(0.0055)	(0.0044)	(0.0024)	(0.0026)	(0.0057)
JSY coverage >50%	0.063***	0.082***	0.10***	0.027***	0.043***	0.045***	0.010
	(0.0081)	(0.0084)	(0.0084)	(0.0057)	(0.0046)	(0.0037)	(0.0073)
Mean of dep. variable at baseline	0.46	0.39	0.20	0.14	0.02	0.02	0.45
Number of observations	342,875	342,875	342,875	342,875	342,875	342,875	340,323

Boxed = the significant >50% coefficients with full controls (Panel B). The effect appears **only at >50%**

The rise is in **community and primary centres** (cols 5–6), not hospitals. ANC (col 7) is flat and unboxed — a falsification check.

Mortality (§5.1, Table 4)

Table 4
Association of JSY with neonatal mortality.

Dependent variable:	Neonatal mortality	Disaggregated measures of mortality		
	(1)	1 day mortality (2)	Death between 2 and 28 days (3)	Death between 8 and 28 days (4)
Panel A. Baseline model				
JSY coverage 10–25%	−0.00078	−0.0013	0.00051	0.00026
	(0.0012)	(0.00085)	(0.00085)	(0.00049)
JSY coverage 25–50%	−0.00030	−0.00048	0.00018	0.000067
	(0.0013)	(0.00093)	(0.00092)	(0.00051)
JSY coverage >50%	−0.0031*	−0.0020*	−0.0011	−0.00057
	(0.0016)	(0.0012)	(0.0011)	(0.00065)
Panel B. Baseline model with district and individual controls				
JSY coverage 10–25%	−0.00043	−0.0012	0.00075	0.00036
	(0.0012)	(0.00086)	(0.00084)	(0.00049)
JSY coverage 25–50%	0.00026	−0.00051	0.00077	0.00030
	(0.0012)	(0.00095)	(0.00089)	(0.00053)
JSY coverage >50%	−0.0027	−0.0022*	−0.00053	−0.00029
	(0.0017)	(0.0012)	(0.0011)	(0.00066)

- Grey = the headline result: neonatal mortality $-2.7/1000$, **not significant** (base 31)
- Amber = one-day deaths -2.2, significant at the **10% level only**
- With full controls, the 95% CI rules out a fall larger than 5.9/1000 — the size *Lim et al.* reported

The correct reading is "no measurable effect", not "no effect".

Slido poll • respond on Slido

Facility births rose, but newborn deaths did not fall. Why, according to the authors?

A The extra births went to low-level centres that cannot handle complications

B The rise in facility use was simply too small to move mortality

C Many "new" facility births were just women switching from private care

D The women drawn in were low-risk, so there were few deaths to prevent

The authors' explanation for the mortality result

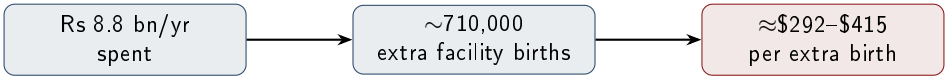
The authors call this *"a persuasive explanation"*:

1. The rise in use was modest (8 pp on a 39% base)
2. The programme sent women **predominantly to facilities below the district hospital** — primary and community centres (Table 3, cols 5–6)
3. The purpose of those facilities *"was not to manage life-threatening complications"*

In the authors' words (§5.1)

"However good the quality of care in health institutions below the district hospital, it may remain inadequate to save the lives of women and [newborns]."

Cost-effectiveness (§5.2)



Caveat 1

The cash goes to **all** facility mothers, including those who would have used a facility anyway. So the marginal cost is understated.

Caveat 2

The spending bought facility *attendance*, not the *survival* that was its goal.

Slido poll • select all you expect

A per-birth cash payment can have effects nobody targeted. Which did the paper find?

A Women switch from private to public facilities

B A small rise in the number of pregnancies

C Earlier breastfeeding after birth

D More caesarean sections (more medical intervention)

Unintended consequences (§5.3, Table 5)

Table 5
Association of JSY with unintended outcomes.

Dependent variable	Place of delivery		Pregnant (2004–2008)	Breastfeeding	
	Private health facility (1)	Public health facility (2)	(3)	Within 1 h (4)	Within 24 h (5)
Panel A. Baseline model					
JSY coverage 10–25%	−0.0053	−0.00072	0.00058	0.016**	0.015*
	(0.0041)	(0.0049)	(0.0011)	(0.0071)	(0.0082)
JSY coverage 25–50%	−0.012***	0.019***	0.0011	0.026***	0.025***
	(0.0042)	(0.0055)	(0.0012)	(0.0085)	(0.0096)
JSY coverage >50%	−0.034***	0.11***	0.0070***	0.075***	0.076***
	(0.0047)	(0.0086)	(0.0019)	(0.011)	(0.013)
Panel B. Baseline model with district and individual controls					
JSY coverage 10–25%	−0.0018	−0.0015	0.0016	0.017**	0.015*
	(0.0039)	(0.0047)	(0.0010)	(0.0072)	(0.0081)
JSY coverage 25–50%	−0.0053	0.013***	0.0026**	0.026***	0.023***
	(0.0040)	(0.0055)	(0.0012)	(0.0087)	(0.0097)
JSY coverage >50%	−0.022***	0.10***	0.0094***	0.073***	0.069***
	(0.0045)	(0.0084)	(0.0020)	(0.012)	(0.014)

Boxed = the significant >50% effects (Panel B):

- **Substitution** (private −2.2 **pp**): some new public births are switchers
- **Fertility** (pregnant +0.9 **pp**): payment per birth can raise births
- **Information** (breastfeeding +7.3 **pp** within 1h, +6.9 **pp** within 24h)

Public facility (+10 pp) is left unboxed: it restates the main result from Table 3.

The fertility result, in the paper's terms

What the authors say. The JSY rewards a birth, so it may raise fertility, through a shorter birth interval or more children over a lifetime. This can work against the mortality goal (§2.2, §5.3).

The design safeguard. A two-child condition on the cash applied **only in low-focus states** and was *"difficult to implement"*. DLHS-3 data show the cash reached all parity groups equally (fn. 24).

The authors' own caveat (fn. 23)

For a woman pregnant at interview, the survey has no start date. The authors **assign** the pregnancy a random month (3–9) and one of two years.

Heterogeneity by person (§6, Table 6)

Table 6
Heterogeneity in the effect of the JSY on government facility births.

	Education of mother		Wealth of household		Residence (in high focus states)		Focal states	
	No education (1)	Some education (2)	Poorest half (3)	Richest half (4)	Urban (5)	Rural (6)	High focus (7)	Low focus (8)
Panel A. Baseline model								
JSY coverage	0.22***	0.18***	0.22***	0.18***	0.19***	0.24***	0.22***	0.0083
	(0.016)	(0.014)	(0.017)	(0.015)	(0.027)	(0.017)	(0.017)	(0.020)
Panel B. Baseline model with district and individual controls								
JSY coverage	0.22***	0.18***	0.23***	0.18***	0.18***	0.24***	0.23***	−0.012
	(0.016)	(0.014)	(0.016)	(0.015)	(0.027)	(0.017)	(0.016)	(0.021)
Mean of dep. variable at baseline	0.11	0.28	0.12	0.28	0.28	0.13	0.16	0.28

The authors report that the effect on facility use is:

- **larger for women with no education** than for women with some education (cols 1–2; $p < 0.001$)
- **larger for poorer women** than for richer women (cols 3–4)

They state that the differences between the subgroups are large relative to the baseline mean.

Slido poll • respond on Slido

The cash is *more generous* in low-focus states. Where would you expect the bigger response?

A Low-focus states — the bigger cash buys a bigger response

B High-focus states — where the supply system (ASHAs, facilities) was also built up

C Whichever started with *lower* facility-birth rates — more room to grow

This one is a genuine puzzle — hold your answer.

Heterogeneity by amount of cash (§6, Table 6)

The authors use the state split to study **one policy parameter**: how the amount of cash affects the response. The cash is larger in rural areas than urban, and larger in low-focus states than high-focus states.

They report that the effect was:

- **larger in rural areas** than urban ($p = 0.058$)
- **larger in low-focus states** than high-focus states ($p < 0.001$) — that is, where the cash is more generous

Read the table with care

The authors' text states the effect is larger in low-focus states. The Table 6 point estimates are high-focus $+\mathbf{0.22}$ and low-focus -0.01 (n.s.). State the text and the table as the paper reports them.

Robustness (§5.4)

The result holds across

- Three-year long differences
- Dropping high-mortality districts
- Dropping low-priority states
- Binary and continuous coverage
- A pre-trend test that is not rejected

State-specific time trends (Panel D)

The authors state this *"reduces the magnitude of the estimates although the general pattern of results remains unchanged"*. The >50% effect falls from **10 pp to 3 pp**.

Back to the opening theme

We opened with the **Ujjwala (LPG)** case: access rose, but the outcome did not follow. The JSY results fit the same pattern.

	Ujjwala (Gupta et al.)	**JSY (this paper)**
Input that rose	LPG ownership ($\approx$75%)	Facility births (+8 pp)
Intended outcome	Clean-air use	Newborn survival
Did it follow?	No — solid fuel persists	No — mortality flat

The next two slides ask *why* the JSY outcome did not follow, using other course readings.

Link: why the supply side matters (Week 3)

The JSY result turns on the **quality of public care**. Our Week 3 readings document exactly that problem:

- **Banerjee, Deaton & Duflo (2004)**, rural Rajasthan: public staff are often **absent**, and people bypass public facilities
- **Das & Hammer (2014)**: low-income settings show a large **"know–do" gap** — providers know more than they do in practice
- **Das et al. (2020), "Two Indias"**: primary-care **quality is low**, especially in the public sector

The connection

These papers give the **mechanism** the JSY study asserts but does not measure: the facilities women were paid to use may not deliver life-saving care. The supply-side constraint is the course's recurring theme.

Link: incentives and the fertility channel

Demand-side incentives (Week 2)

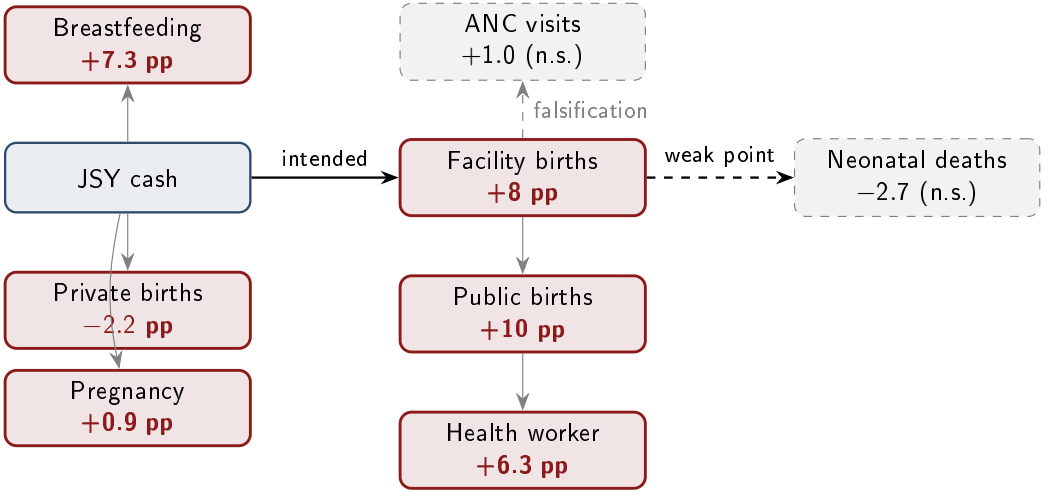
Dupas (2011), "Health Behaviour in Developing Countries", reviews how prices and subsidies shape health demand. The JSY is a textbook demand-side price change, and fits the finding that demand responds but **health gains are not guaranteed**.

Fertility as a response (Weeks 4, 6)

The JSY's **fertility side effect** connects to the fertility unit: **Dehejia & Lleras-Muney (2004)** on economic conditions and births, and **Anukriti et al.** on the quantity–quality trade-off. A per-birth payment is an economic shock to fertility.

A further contrast: **RSBY** health insurance (Karan et al. 2017; Prinja et al. 2019, Week 10) is another large Indian demand-side scheme with weak effects on the intended outcome.

All results on one map



Red = significant, grey dashed = not significant. A summary of the results in Tables 3, 4 and 5.

Limitations stated by the authors

The authors state these limitations in the paper.

- **Endogeneity.** The authors write that they *"cannot rule out"* confounding. Districts may have adopted the JSY early because of other changes (§1, §4)
- **The assumption cannot be tested.** It is *"by definition impossible to test"* the parallel-trends assumption formally (§5.4)
- **The treatment measure.** JSY coverage is *"mechanically linked to the fraction of women giving birth in a facility"*, so it is an imperfect measure of treatment (§2.3)
- **Imprecise mortality estimate.** The confidence intervals are *"not sufficiently tight to reject modest effects"* on mortality (§5.1, §7)
- **The pregnancy measure.** For a woman pregnant at interview, the authors **impute** the date of the pregnancy (fn. 23)
- **External validity.** Extending the estimates past 2008 *"may not provide a good approximation to the true impact"* (§7)

Future research, as the authors propose it (§7)

In the conclusion, the authors give these directions for future work.

- Study the **long-term effects** of financial-incentive programmes, not only the short-term effects
- Study the **potential harms** that such programmes can cause (Lagarde et al., 2007)

The authors' main policy point

The evidence on intended and unintended effects points to the need for **caution** in the use of financial incentives. The supply side must be in place for a demand-side incentive to work, but many schemes run where the quality and availability of care are inadequate.

The authors' conclusions (§7)

In their own words, the authors conclude:

1. The JSY is associated with an increase in the use of formal health care, **particularly at lower levels** of the public health system. The increase was **modest**
2. The findings on neonatal mortality show **no strong evidence of an effect**, though the confidence intervals cannot reject modest effects
3. The evidence on intended and unintended effects points to the need for **caution** in the use of financial incentives
4. The supply side **must be in place** for a demand-side incentive to work

Thank you

Questions?

Backup: comparison with Lim et al. (2010)

Outcome	Lim et al. (2010)	This paper (>50%)
Facility births	+44 to 49 pp	+8.2 pp
ANC (3+ visits)	+11 pp	+1.0 pp (n.s.)
Neonatal mortality	−2 to −6/1000	−2.7/1000 (n.s.)

The Lim et al. mortality effect is significant in only two of three methods.

Backup: state-specific trends (Table A7)

Public facility delivery	Baseline (Table 3)	State trends (A7)
Coverage 10–25%	−0.2 pp	−1.0 pp**
Coverage 25–50%	+1.3 pp**	−1.0 pp**
Coverage >50%	+10.0 pp***	+3.3 pp***

Backup: pregnancy measurement

- DLHS-3 only. The recall is truncated at 1 April 2001 (2004 for DLHS-3)
- For a woman pregnant at interview, the start month is random (3 to 9) and the year is one of two
- Table 5, col 3 uses woman-years, so the unit differs from the other columns